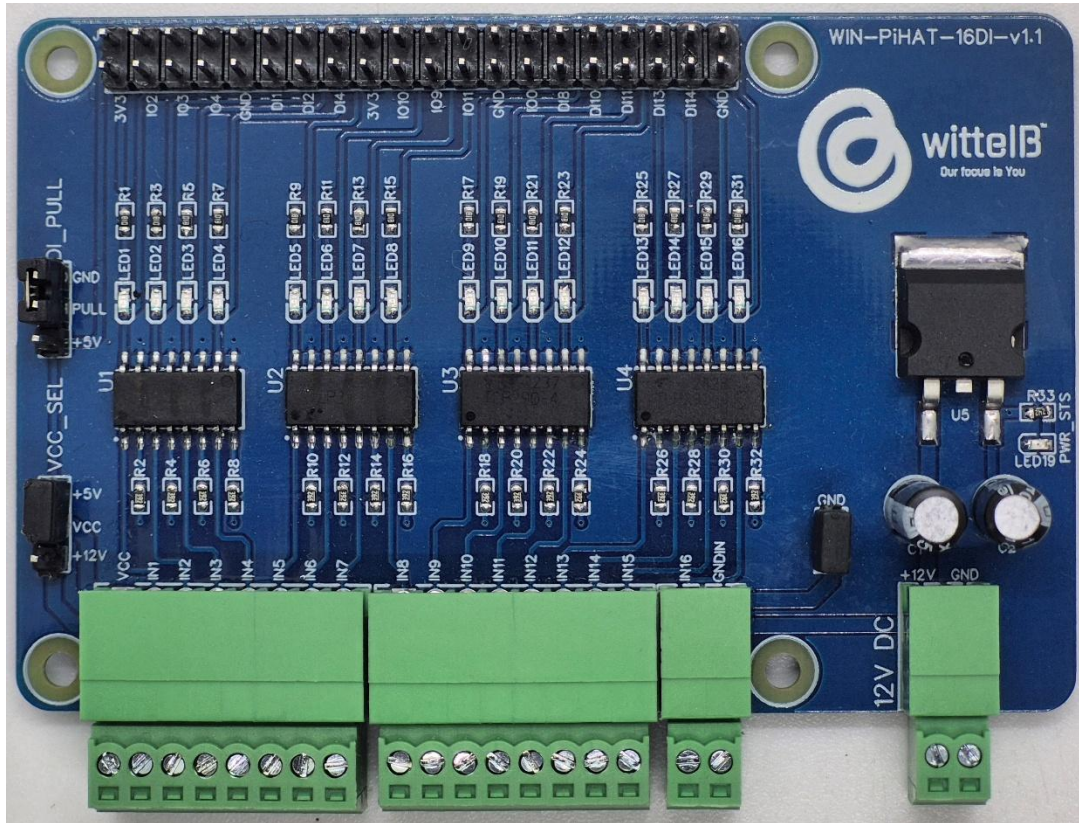


Technical Datasheet of WIN-PiHAT-16DI

1. Overview

The **Digital Input (DI) PiHAT** is a 16-channel expansion board for Raspberry Pi 4 and CM4, offering **opto-isolated signal** acquisition. It protects your system by providing electrical isolation between external digital inputs and the Pi's GPIO header, ensuring high signal integrity for industrial environments.



PiHAT features a scalable design for 4, 8, or 16 channels(DI). This flexible, population-based assembly makes it a compact and reliable solution for embedded control, monitoring, and automated industrial applications.

Specifications

Category	16DI PiHAT
Input Type	Digital Input (12V & 5V with Reference), Digital Input (With GND Reference), Digital Input (80V Absolute Max. @50mA)
Channel Count	16 Channels (Expandable Design Architecture)
Connectors	2 of 2.54 mm Pluggable Terminal Blocks (8-Pin), 2 of 2.54 mm Pluggable Block (2-Pin), 40-Pin Raspberry Pi GPIO Header

General :-

Dimensions	56 mm L x 85 mm W
Power	Input Power – 12/24V DC
Operating Temperature	-30 – 80° C (-22 ~ 176°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DI (Digital Inputs) :-

Digital Input channels	16
Sense Voltage	3.3 – 12V DC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Additional Features :-

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, ± 30 KV contact, ± 30 KV air

Configuration :-

All 16DI channels operate as opto-isolated digital inputs and are directly interfaced to the Raspberry Pi GPIO header.

Input configuration and signal handling are performed through software running on the Raspberry Pi. Each input channel is initialized as a GPIO input and can be monitored using standard Raspberry Pi GPIO libraries. The input logic is active-low, with the optocoupler output pulling the corresponding GPIO pin to ground when the input signal is present.

The following are step by step guides for configuration:

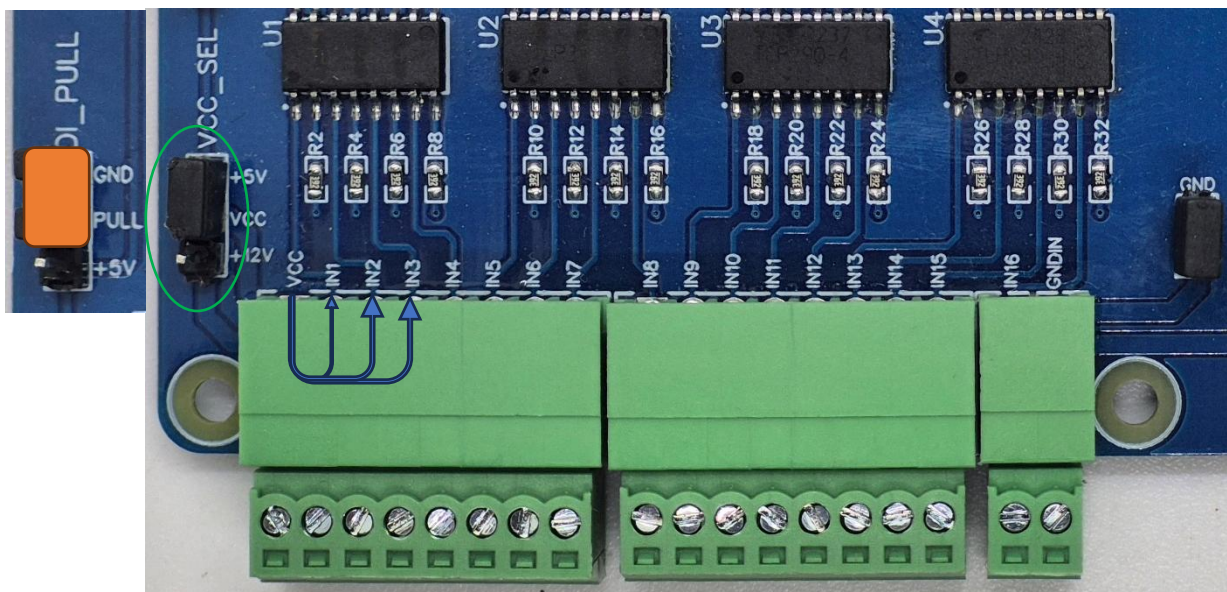
1. Connect the Module to +12/24V DC POWER SUPPLY.
2. Connect the Board to the Raspberry Pi via header pins

Sr. No.	Digital Input	Pi GPIO	Header Pin number	Alternate Function
1	DI1	17	11	-
2	DI2	27	13	-
3	DI3	23	16	-
4	DI4	22	15	-
5	DI5	24	18	-
6	DI6	25	22	-
7	DI7	07	26	CE1
8	DI8	05	29	-
9	DI9	12	32	PWM0
10	DI10	06	31	-
11	DI11	13	33	PWM1
12	DI12	16	36	-
13	DI13	19	35	PCM FS
14	DI14	26	37	-
15	DI15	20	38	PCM DIN
16	DI16	21	40	PCM DOUT

Note: Using any listed **Alternate Function** will **disable** that specific channel's ability to act as a **Digital Input**, as these pins can only **support one hardware mode** at a time.

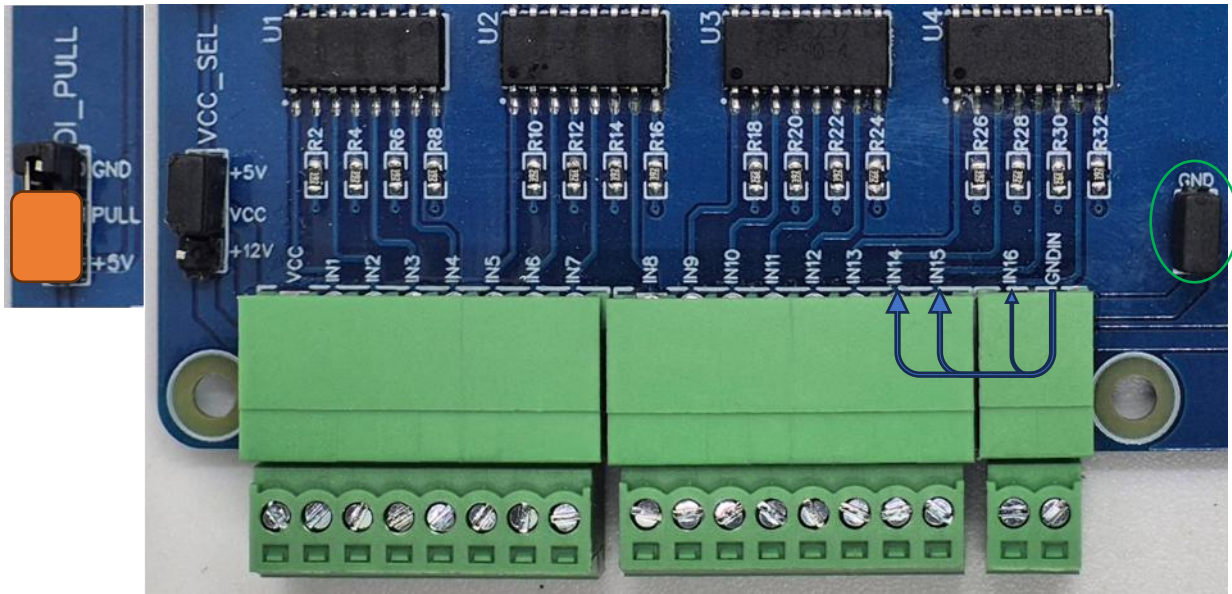
Wiring Diagram:-

1. For 5 & 12V Reference:



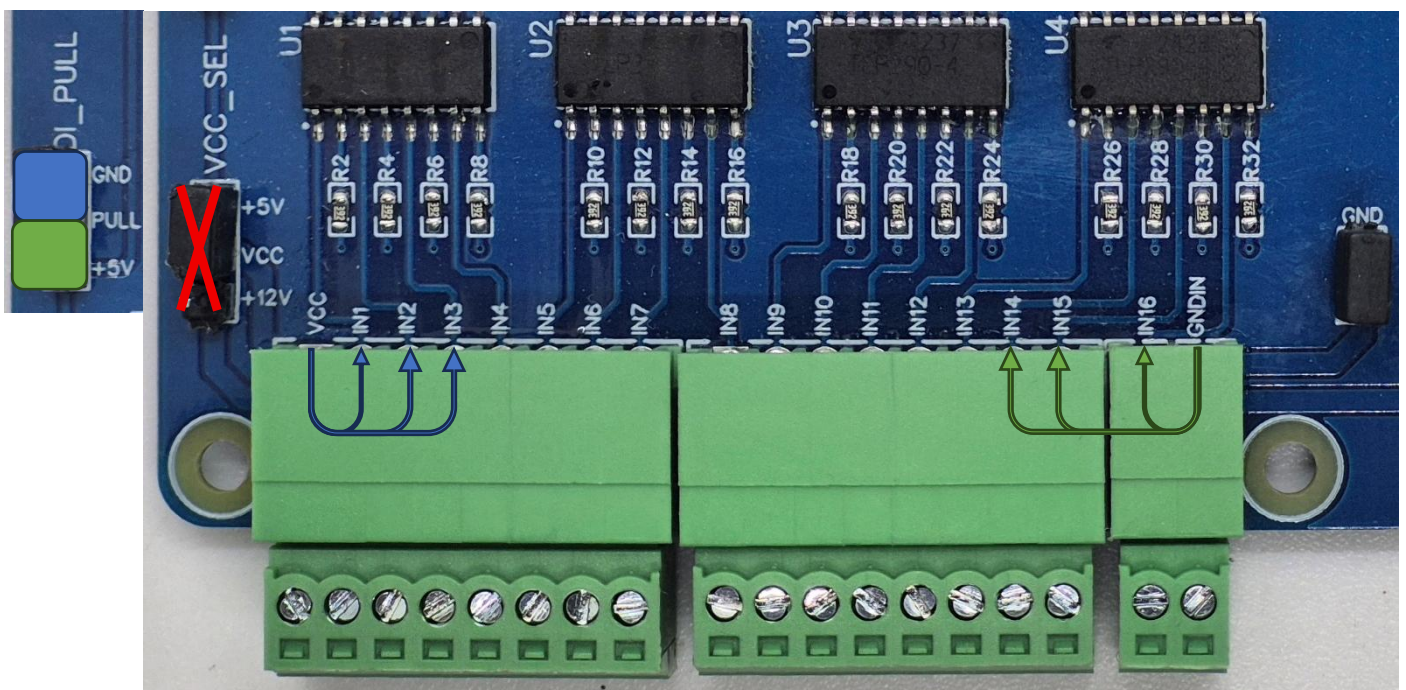
For Digital Input with 12V or 5V reference, VCC_SEL jumper should be connected to +12V or +5V. And DI_PULL jumper should be connected to GND.

For Ground Reference:



For Digital Input with Ground reference, GND jumper should be connected. And DI_PULL jumper should be connected to +5V.

2. For External Reference Voltage:

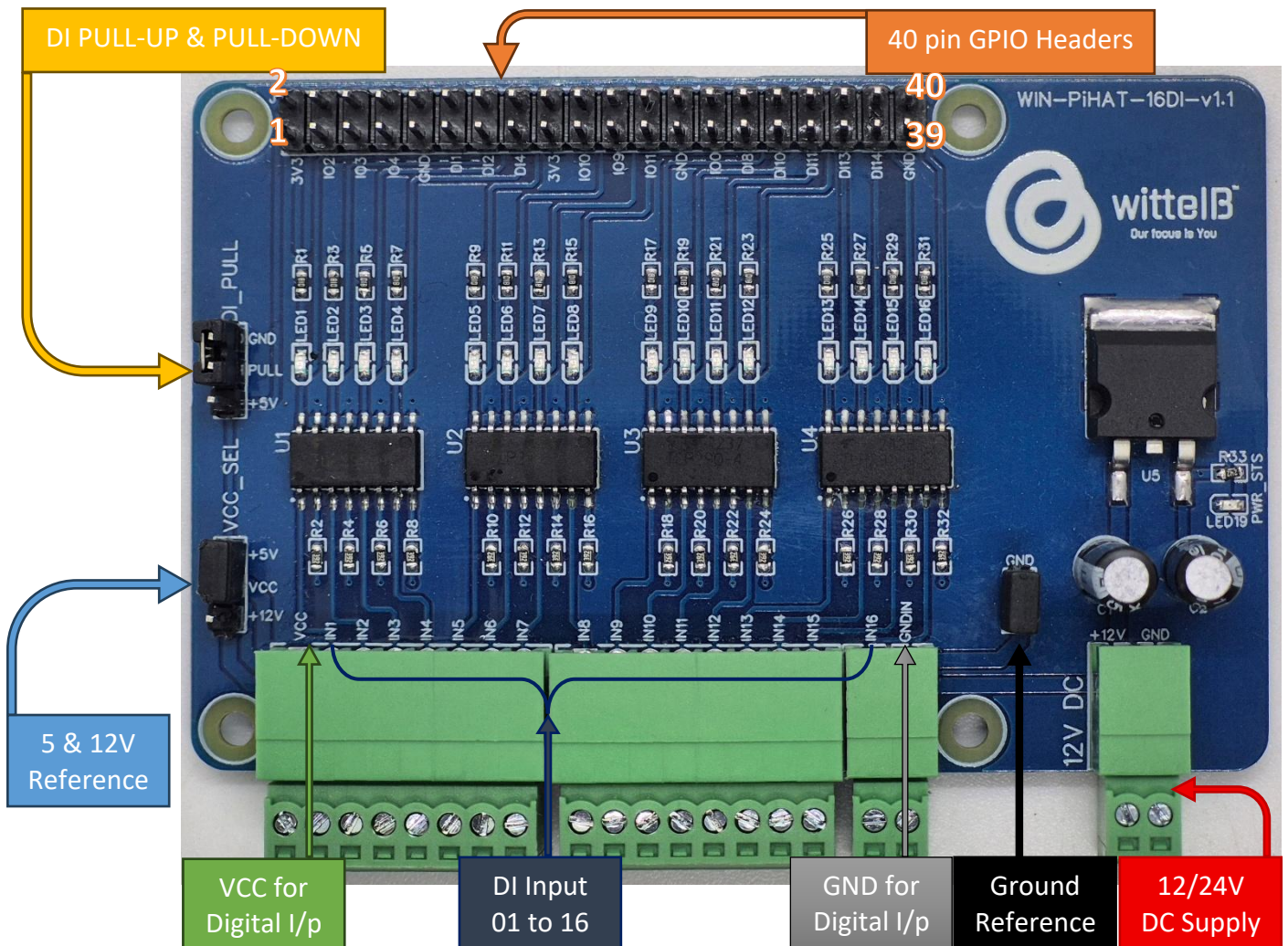


For Digital Input with External Supply, VCC_SEL and GND jumper are Disconnected. And as per requirement if DI is with pull-up then DI_PULL jumper is connected to +5V and DI is connected to GNDIN. And if DI is with pull-down then DI_PULL jumper is connected to GND and the DI is connected to VCC.

Sample Test Code:

- [WIN-PiHAT-16DI](#)

Hardware Interface & Connector Layout:



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